```
n1 = 5;
g1 = 3;
n2 = 5;
g2 = 1;
n3 = 5;
g3 = 1;


(*Consistencies solution for the choice of pentaquark *)

γ12bar = 0.3987206213820558; (* relationship between light and anti-charm quark *)
β12 = 0.21393789137970892; (* relationship betwween light and charm quark *)
β22bar = 0.824988803688124; (* relationship between charm and anti-charm quark *)

(*Defining Constants*)

xx = 0.346;
(*This is the value of αs which is αs=g^2/(ħc),
used from the mesonic fit as done in meson calculations.*)
alfas = xx;
(*aa=.178;*)
 zz = 306.;(*This is the mass of light quark like up, down*)
 aa = 0.197; (*This is the ratio of light mass to the heavy mass of quarks *)
yy = 107.6;
B0 = yy; (*Bag Constant *)
a1 = 1 / zz; (*This is α=ħ/(m1 c), where m1 is the mass of light quark. here,
reduced planck's constant and c both are set to be 1.*)
P = N[Pi, 7];
alfa1 = 1.;(* Ratio of mass of light quarks, BA, 13 September *)
alfa2 = aa; (* Ratio of mass of light quark to the heavy quark,
BA, 13 September *)

η = (n1 * g1 + n3 * g3 + n2 * g2)/5;
theta = -6 η/(5 (5 η - 1)); (*Just a multiplication factor for a particular value of eta *)
Q1 = -0.525; (*Q1 is the constant in the differential equation  d^2 f1/dx^2 =
```

Q1 $\frac{f1^{\frac{3}{2}}}{\sqrt{x}}$ for the outer region*)

```
Q3 = -0.9585131676908107;
(*Q3 is the constant in the differential equation  d^2 f3/dx^2 =
 Q3 f3^(3/2)/Sqrt[x] in the middle region. This is obtained by solving two
  consistency conditions along with other constant γ12bar. *)
Q2 = -1.9948905987395598 ;
(*Qbar is the constant in the differential equation  d^2 f3/dx^2 =
 Q2 f3^(3/2)/Sqrt[x] in the inner region. This is obtained by solving three
    consistency conditions along with other constants β13 & β23. *)
g = alfas;(*Why  g = alfas in this case ?*)
(* I think there should be g = /sqrt{alfas} in order to satisfy the multiplication
  facor in the expreesion of kinetic energy i.e. 'c', BA, 13 September *)
R = 3 a/(8 * alfas) * (3 π η/2)^(2/3);(*Page number 54, it is inverse in the copy though *)
c = 12/(5 π) 4/3 g^2 - 4/3 alfas/a (3 π η/2)^(1/3) ;
(*Multiplication factor in the expression of kinetic energy *)
d = 4/3 g^2 - 4/3 alfas/a (3 π η/2)^(1/3) 18/(25 π) η/(η - 1/5);  (* 18g^2/(5(5η -1 )).16 π^2.(3a/8alfas). R^2/(6π^2)^2 ,
The value of R is from page number 54 which is written inverse here,
and (3a/8alfas) is inverse in copy, BA, 13 September *)
(*Multiplication factor in the expression of Potential energy *)


(*Let us define the three functions for three regions *)

f3[x3_?NumericQ, x2_?NumericQ, u2_?NumericQ , s2_?NumericQ] :=
  Abs[x3 * y3'[x3] - y3[x3] - n3 * Q3/(3 η) - β22bar * n2 * Q2/(3 η) + β22bar^(3/2) * n2 * Q3/(3 η)]
```

```
(*Appendix from Kinetic energy*) /.
 First@NDSolve[{y3''[x] == (Q3 * Abs[y3[x]^(3/2)] / x^(1/2)), y3'[x2] == β22bar * s2 ,
    y3[x2] == β22bar * u2 (* It shows how the function f3
     (x3) changes on other regions*)}, y3, {x, x2, x3}];
(* u2 is the function f_2(x) *)
(* s2 is the first derivative of function f_2(x) *)
(*This is the function f3 defined on the
  middle region. It has the differential equation  d^2 f3/dx^2 =
 Q3 f3^(3/2)/Sqrt[x] in for x3<x<x2 which is middle region. On this region, f1(x)=γ12 f2(x).  *)
f1[x1_?NumericQ, x3_?NumericQ, s3_?NumericQ, u3_?NumericQ] :=
  Abs[x1 * y1'[x1] - y1[x1] - n1 * Q1/(3 η) - γ12bar * n3 * Q3/(3 η) +
    γ12bar^(3/2) * n3 * Q1/(3 η) - (γ12bar^(3/2) * β22bar^(3/2) - β12^(3/2)) * n2 * Q1/(3 η) +
    γ12bar * β22bar^(3/2) * n2 * Q3/(3 η) - γ12bar * β22bar * n2 * Q2/(3 η)]
 (*Appendix from Kinetic energy*) /. First@NDSolve[{y1''[x] == (Q1 * Abs[y1[x]^(3/2)] / x^(1/2)),
    y1'[x3] == γ12bar * s3, (* s3 is the first derivative of function f_2bar(x) *)
    (* It means at the boundary x3 the function df1 changes to other function *)
    y1[x3] == γ12bar * u3 (* u3 is the function f_2bar(x) *)
    (*It means at the boundary x3 the function f1 changes to other function*)
   }, (* It means at the boundary x3 the function f1 changes to other function *)
   y1, {x, x3, x1}]; (*This is the function defined in outer region for x2<
  x<x1 so called f1. It has differential equation  d^2 f1/dx^2 =
 Q1 f1^(3/2)/Sqrt[x] for the outer region which is solved by using boundary
   condition of (d/dx) f1(x)|x2 = γ12bar (d/dx) f2(x)|x2 and f1(x3)=γ12 f2(x3)  *)
```

```
f2[xmax_?NumericQ, m2_?NumericQ] :=
  Module[{xx = xmax},
   solution2 = NDSolve[
     {y2''[x] == (Q2 * Abs[y2[x]^3/2]) / x^1/2, y2'[1.*^-9] == m2, y2[1.*^-9] == 0.}, y2, {x, 1.*^-9, xx}];

   m = NIntegrate[Sqrt[x] * Evaluate[y2[x] /. solution2]^(3 / 2),
     {x, 1.*^-9, xx}, Method → {Automatic, "SymbolicProcessing" → 0}];
   z = m[[1]];
   z = z - n2 / (3. * η);
   (*z=z-n/(A)*);
   Return[Abs[z]]; Remove["NDSolve`*$*$1"]];
(*This is the function f3 defined on the inner region. It has the differential
  equation  d^2 f3/dx^2 = Q3 f3^(3/2)/Sqrt[x]  for 0<x<x_3 inner region. On this region,
f_1(x)=β_13 f_3(x) and f_2(x)= β_23 f_3(x).   *)

(*Start the program*)

m2 = 32;
For[j = 1., j ≤ 1., j++, B = (B0)^4;
  a = a1; (* It must be to satisfy the multiplication
   factor used in kinetic energy i.e. c, BA, 13 September *)
  (*Now for the first pentaquark,
  eta=1 which in total contains 5 quarks which can be written by *)

 mqn = If[j > 1, m2 + 2, m2]; (*case2new*)(*mqn=If[j>1,m2+4,m2];*)
  m2 = mqn;
  topi = 0.01;
  m2B = m2;
  Energy = 10^9;
  Energy = 10^9;
  Energy2plus = 10^9;
```

```
H = 10^9 + 1;
SN = 0;
nloop = 1;
For[n = 1, n ≤ 3, n++, While[Energy ≤ H, F = H;
  H = Energy;
  HX = Energy6bare;
  HY = Energy2plus; m2 = m2 + SN * 10^(-n);
  x2 = 2.*^-8; (*Print[m2];*)(*was 0 .5 1.66*)
  FindMinimum[f2[x2, m2], {x2, topi}];
  (*remove this function and investigate if it is actually required BS 4/3/2021*)
  x2 = x2 /. Last[FindMinimum[f2[x2, m2], {x2, topi}]];
  solution2 = NDSolve[{y2''[x] == (Q2 * Abs[y2[x]^(3/2)]) / x^(1/2),
     y2'[1.*^-9] == m2, y2[1.*^-9] == 0.}, y2, {x, 1.*^-9, x2}];
  t1 = Evaluate[y2[x2] /. solution2];
  t4 = Evaluate[y2'[x2] /. solution2];
  u2 = t1[[1]]; (* u2 is the function f_2(x) *)
  s2 = t4[[1]]; (* s2 is the first derivative of function f_2(x) *)
  If[u2 < 0., u2 = 0.];


  x3 = x2 + 0.1;
  FindMinimum[{f3[x3, x2, u2, s2]}, {x3, x2}];
  x3 = x3 /. Last[FindMinimum[{f3[x3, x2, u2, s2]}, {x3, x2}]];
  solution3 = NDSolve[{y3''[x] == (Q3 * Abs[y3[x]^(3/2)]) / x^(1/2),
     y3'[x2] == β22bar * s2 , y3[x2] == β22bar * u2}, y3, {x, x2, x3}];
  t2 = Evaluate[y3[x3] /. solution3];
  t5 = Evaluate[y3'[x3] /. solution3];
  u3 = t2[[1]]; (* u3 is the function f_2bar(x) *)
  s3 = t5[[1]]; (* s3 is the first derivative of function f_2bar(x) *)
  If[u3 < 0., u3 = 0.];

  x1 = x3 + 0.1;
  FindMinimum[{f1[x1, x3, s3, u3]}, {x1, x3}];
  x1 = x1 /. Last[FindMinimum[{f1[x1, x3, s3, u3]}, {x1, x3}]];
```

```
solution1 = NDSolve[{y1''[x] == (Q1 * Abs[y1[x]^(3/2)] / x^(1/2)),
    y1'[x3] == γ12bar * s3 , y1[x3] == γ12bar * u3 }, y1, {x, x3, x1}];
t3 = Evaluate[y1[x1] /. solution1];
t6 = Evaluate[y1'[x1] /. solution1];
u1 = t3[[1]]; (* u1 is the function f_1(x) *)
s1 = t6[[1]]; (* s1 is the first derivative of function f_1(x) *)
If[u1 < 0., u1 = 0.];


(*Integrations *)(* These integrations change values but outer and outer 1
 doesnot because they depend on x and derivates of x (BA, 11 September)*)

normmx2 = x2 s2 - u2 ;
normx2 = n2 Q2 / (3 η) ;
(*normx2barr and normx2bar should be equal *)
normmx3 = x3 s3 - u3 ;
normx3 = n3 Q3 / (3 η) + β22bar n2 Q2 / (3 η) - (β22bar)^(3/2) n2 Q3 / (3 η);
(*normmx2 and normx2 should be equal *)
normmx1 = x1 s1 - u1 ;
normx1 = n1 Q1 / (3 η) + γ12bar n3 Q3 / (3 η) - (γ12bar)^(3/2) n3 Q1 / (3 η) +
  ((γ12bar)^(3/2) (β22bar)^(3/2) - (β12)^(3/2)) n2 Q1 / (3 η) - γ12bar (β22bar)^(3/2) n2 Q3 / (3 η) + γ12bar β22bar n2 Q2 / (3 η) ;
(*normmx1 and normx1 should be equal *)

h1 = NIntegrate[Evaluate[y2[x] /. solution2]^2.5 / (x^0.5),
  {x, 0, x2}, Method → {Automatic, "SymbolicProcessing" → 0}];
h2 = NIntegrate[Evaluate[y3[x] /. solution3]^2.5 / (x^0.5), {x, x2, x3},
  Method → {Automatic, "SymbolicProcessing" → 0}];
h3 = NIntegrate[Evaluate[y1[x] /. solution1]^2.5 / (x^0.5), {x, x3, x1},
  Method → {Automatic, "SymbolicProcessing" → 0}];
h4 = h1[[1]];
h5 = h2[[1]];
```

```
h6 = h3[[1]] ;

AB = 4/7 Sqrt[x2] (u2)^(5/2) - 5 n2/(21 η) s2 ;

CD = 4/7 Sqrt[x3] (u3)^(5/2) - 4/7 (β22bar)^(5/2) Sqrt[x2] (u2)^(5/2) -
   5/(7 Q3) s3 (n3 Q3/(3 η) + β22bar n2 Q2/(3 η) - (β22bar)^(3/2) n2 Q3/(3 η)) + 5/(7 Q3) (β22bar)^2 n2 Q2/(3 η) s2 ;

CCD = 4/7 Sqrt[x3] (u3)^(5/2) - 4/7 Sqrt[x2] (β22bar)^(5/2) (u2)^(5/2) - 5/(7 Q3) s3 normmx3 +
   5/(7 Q3) (β22bar)^2 normmx2 s2 ;

EF = 4/7 Sqrt[x1] (u1)^(5/2) - 4/7 (γ12bar)^(5/2) Sqrt[x3] (u3)^(5/2) -
   5/(7 Q1) s1 (n1 Q1/(3 η) + γ12bar n3 Q3/(3 η) - (γ12bar)^(3/2) n3 Q1/(3 η) + n2 Q1/(3 η) ((γ12bar)^(3/2) (β22bar)^(3/2) - (β12)^(3/2)) -
      (γ12bar) (β22bar)^(3/2) n2 Q3/(3 η) + (γ12bar) (β22bar) n2 Q2/(3 η)) +
   5/(7 Q1) (γ12bar)^2 s3 (n3 Q3/(3 η) + β22bar n2 Q2/(3 η) - (β22bar)^(3/2) n2 Q3/(3 η)) ;
(* h4 and AB should be equal *)
(* h5 and CD should be equal *)
(* h6 and EF should be equal *)


K222 = n2/(3 η Q2) s2 - 1/Q2 h4 ;

K22 = 4 n2/(7 η Q2) s2 - 4/(7 Q2) Sqrt[x2] (u2)^(5/2) ;

(*K2bar2barr and K2bar2bar should be equal *)

K32 = - 4/(7 Q2) β22bar^1.5 Sqrt[x2] (u2)^(5/2) +
```

$$\frac{n2}{3\,\eta}\left(\frac{\beta 22bar^{1.5}}{Q2}-\frac{\beta 22bar}{Q3}+\frac{5*\beta 22bar^{1.5}}{7\,Q2}\right)s2+\frac{n2}{3\,\eta\,Q3}\,s3\,;$$

$$K332=\frac{n2\,(\beta 22bar)^{\frac{3}{2}}}{3\,\eta\,Q2}\,s2-\frac{n2\,\beta 22bar}{3\,\eta\,Q3}\,s2+\frac{n2}{3\,\eta\,Q3}\,s3-\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2}\,h4;$$

(*K2bar2 and K2barr2 should be equal *)

$$K21=-\frac{4}{7\,Q2}\,\beta 12^{1.5}\sqrt{x2}\;(u2)^{\frac{5}{2}}+\frac{n2}{3\,\eta}\left(\frac{5}{7\,Q2}\,\beta 12^{1.5}+\frac{\beta 12^{1.5}}{Q2}-\frac{\gamma 12bar^{1.5}*\beta 22bar}{Q3}\right)s2+$$

$$\frac{n2}{3\,\eta}\left(\frac{\gamma 12bar^{1.5}}{Q3}-\frac{\gamma 12bar}{Q1}\right)s3+\frac{n2}{3\,\eta\,Q1}\,s1\,;$$

$$K221=\frac{n2}{3\,\eta\,Q2}\,(\beta 12)^{\frac{3}{2}}\,s2+\frac{n2\,(\gamma 12bar)^{\frac{3}{2}}}{3\,\eta\,Q3}\,s3-$$

$$\frac{n2}{3\,\eta\,Q3}\,(\gamma 12bar)^{\frac{3}{2}}\,\beta 22bar\,s2+\frac{n2}{3\,\eta\,Q1}\,s1-\frac{n2}{3\,\eta\,Q1}\,\gamma 12bar\,s3-\frac{(\beta 12)^{\frac{3}{2}}}{Q2}\,h4\,;$$

(*K2bar1 and K2barr1 should be equal *)

$$K31=\frac{4}{7}\,\beta 22bar^{1.5}\left(\frac{\gamma 12bar^{1.5}*\beta 22bar}{Q3}-\frac{\beta 12^{1.5}}{Q2}\right)\sqrt{x2}\;(u2)^{\frac{5}{2}}-\frac{4}{7\,Q3}\,\gamma 12bar^{1.5}\sqrt{x3}\;(u3)^{\frac{5}{2}}+$$

$$\frac{n2}{3\,\eta}\,\beta 22bar\left(\frac{5}{7\,Q2}\,\beta 22bar^{\frac{1}{2}}*\beta 12^{1.5}-\frac{\gamma 12bar^{1.5}*\beta 22bar^{1.5}}{Q3}-\frac{5\,Q2}{7\,Q3^2}\,\gamma 12bar^{1.5}*\right.$$

$$\left.\beta 22bar+\frac{Q2}{Q3^2}\,\gamma 12bar^{1.5}*\beta 22bar+\frac{1}{Q2}\,\beta 12^{1.5}*\beta 22bar^{\frac{1}{2}}-\frac{\beta 12^{1.5}}{Q3}\right)s2+$$

$$\frac{1}{3\,\eta\,Q3}\left(n3*\gamma 12bar^{1.5}-\frac{n3\,Q3}{Q1}\,\gamma 12bar+n2*\beta 12^{1.5}-\frac{n2\,Q2}{Q3}\,\gamma 12bar^{1.5}*\beta 22bar+\right.$$

$$\left.\frac{5}{7}\,\gamma 12bar^{1.5}\left(n3+\frac{n2\,Q2}{Q3}\,\beta 22bar-n2*\beta 22bar^{1.5}\right)\right)s3+\frac{n3}{3\,\eta\,Q1}\,s1\,;$$

$$(*K122 = \frac{n2}{3\,\eta}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}\left[\frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] - \frac{(\beta 12)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}}{Q2}h4 - \frac{(\gamma 12bar)^{\frac{3}{2}}}{Q3}h5$$

$$- \frac{n2\,Q2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\left[\frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] + \frac{n3}{3\,\eta\,Q1}s1 - \frac{n3}{3\,\eta\,Q1}\gamma 12bar\,s3$$

$$+ (\beta 12)^{\frac{3}{2}}\frac{n2}{3\,\eta}\left[\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2}s2 + \frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] +$$

$$\frac{n3}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}s3\ ;\ *)$$

$$K331 = \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}\beta 22bar\,s2 - \frac{(\beta 12)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}}{Q2}h4 - \frac{(\gamma 12bar)^{\frac{3}{2}}}{Q3}h5 -$$

$$\frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s3 + \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^2\,s2 + \frac{n2}{3\,\eta}(\beta 22bar)^{\frac{3}{2}}$$

$$\left(\frac{\beta 12^{\frac{3}{2}}}{Q2} - \frac{(\gamma 12bar)^{\frac{3}{2}}\beta 22bar}{Q3}\right)s2 + \frac{n3}{3\,\eta}\gamma 12bar\left(\frac{\gamma 12bar^{\frac{1}{2}}}{Q3} - \frac{1}{Q1}\right)s3 + \frac{n3}{3\,\eta\,Q1}s1\ ;$$

(*K12 and K122 should be equal*)

$$K33 = \frac{4}{7}\beta 22bar^2\left(\frac{\beta 22bar^{\frac{1}{2}}}{Q3} - \frac{\beta 22bar}{Q2}\right)\sqrt{x2}\,(u2)^{\frac{5}{2}} -$$

$$\frac{4}{7\,Q3}\sqrt{x3}\,(u3)^{\frac{5}{2}} + \frac{n2}{3\,\eta}\beta 22bar^2\left(\beta 22bar\frac{12}{7\,Q2} + \frac{2}{7}\frac{Q2}{Q3^2} - \frac{2*\beta 22bar^{\frac{1}{2}}}{Q3}\right)s2 +$$

$$\left(\frac{12\,n3}{21\,\eta\,Q3} + \beta 22bar\frac{n2}{3\,\eta\,Q3}\left(\beta 22bar^{\frac{1}{2}} - \frac{2}{7}\frac{Q2}{Q3} - \frac{5}{7}\beta 22bar^{\frac{1}{2}}\right)\right)s3\ ;$$

$$K333 = \frac{n2\,Q2}{3\,\eta\,Q3}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2} - \frac{\beta 22bar}{Q3}\right)s3 - \frac{n2\,Q2\,\beta 22bar}{3\,\eta\,Q3}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2} - \frac{\beta 22bar}{Q3}\right)s2 -$$

$$\frac{(\beta 22bar)^3}{Q2}h4 - \frac{1}{Q3}h5 + (\beta 22bar)^3\frac{n2}{3\,\eta\,Q2}s2 - (\beta 22bar)^{\frac{5}{2}}\frac{n2}{3\,\eta\,Q3}s2 + \frac{n3}{3\,\eta\,Q3}s3\ ;$$

(*K22 and K222 should be equal *)

(*THese Y and Z are to check the value of integral of power $\frac{5}{2}$ in the outer region. Which proved to be false yet. I need to work on it extensively.*)

$$K11 = \frac{4}{7}\left(\frac{\beta22bar^{\frac{5}{2}} * \gamma12bar^3}{Q3} - \frac{\beta12^3}{Q2}\right)\sqrt{x2}\ (u2)^{\frac{5}{2}} + \frac{4}{7}\left(\frac{\gamma12bar^{\frac{5}{2}}}{Q1} - \frac{\gamma12bar^3}{Q3}\right)\sqrt{x3}\ (u3)^{\frac{5}{2}} - \frac{4}{7\,Q1}\sqrt{x1}\ (u1)^{\frac{5}{2}} + \left(\frac{n2}{3\,\eta}\left(\frac{\beta12^3}{Q2} - \frac{\beta12^{\frac{3}{2}} * \gamma12bar^{\frac{3}{2}} * \beta22bar}{Q3} + \frac{Q2}{Q3^2}\gamma12bar^3 * \beta22bar^2 - \frac{\beta12^{1.5} * \gamma12bar^{1.5} * \beta22bar}{Q3}\right) + \frac{5\,n2}{21\,\eta}\left(\frac{\beta12^3}{Q2} - \frac{Q2}{Q3^2}\beta22bar^2 * \gamma12bar^3\right)\right)s2 + \left(\frac{n2}{3\,\eta}\left(\frac{\beta12^{1.5} * \gamma12bar^{1.5}}{Q3} - \frac{\beta12^{1.5} * \gamma12bar}{Q1} - \frac{\beta22bar^{1.5} * \gamma12bar^3}{Q3} + \frac{\beta22bar^{1.5} * \gamma12bar^{\frac{5}{2}}}{Q1} + \frac{\beta12^{1.5} * \gamma12bar^{1.5}}{Q3} - \frac{Q2}{Q3^2}\gamma12bar^3 * \beta22bar - \frac{\beta12^{1.5} * \gamma12bar}{Q1} + \frac{\gamma12bar^{\frac{5}{2}} * \beta22bar^{1.5}}{Q1} + \frac{Q2}{Q1^2}\gamma12bar^2 * \beta22bar - \frac{Q3}{Q1^2}\beta22bar^{1.5} * \gamma12bar^2\right) + \frac{n3}{3\,\eta}\left(\frac{\gamma12bar^3}{Q3} - \frac{2 * \gamma12bar^{\frac{5}{2}}}{Q1} + \frac{Q3}{Q1^2}\gamma12bar^2\right) + \frac{5}{21\,\eta}\left(\frac{n3}{Q3}\gamma12bar^3 + \frac{n2\ Q2}{Q3^2}\gamma12bar^3 * \beta22bar - \frac{n2}{Q3}\beta22bar^{1.5} * \gamma12bar^3 - \frac{n3\ Q3}{Q1^2}\gamma12bar^2 - \frac{n2\ Q2}{Q1^2}\gamma12bar^2 * \beta22bar + \frac{n2\ Q3}{Q1^2}\gamma12bar^2 * \beta22bar^{1.5}\right)\right) * s3 + \left(\frac{n1}{3\,\eta\,Q1} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\gamma12bar + \frac{n3}{3\,\eta\,Q1}\gamma12bar^{1.5} + \frac{n2}{3\,\eta\ Q1}\right.$$

$$\left(\beta 12^{1.5} - \gamma 12bar^{1.5} * \beta 22bar^{1.5} - \frac{Q2}{Q1}\beta 22bar * \gamma 12bar + \frac{Q3}{Q1}\beta 22bar^{1.5} * \gamma 12bar\right) + \frac{5}{21\,\eta}\left(\frac{n1}{Q1} + \frac{n3\ Q3}{Q1^2}\gamma 12bar - \frac{n3}{Q1}\gamma 12bar^{1.5} + \frac{n2}{Q1}\gamma 12bar^{1.5} * \beta 22bar^{1.5} - \frac{n2}{Q1}\beta 12^{1.5} - \frac{n2\ Q3}{Q1^2}\beta 22bar^{1.5} * \gamma 12bar + \frac{n2\ Q2}{Q1^2}\gamma 12bar\ \beta 22bar\right)\Bigg) * s1;$$

$$K111 = \frac{-(\beta 12)^3}{Q2}h4 - \frac{(\gamma 12bar)^3}{Q3}h5 - \frac{1}{Q1}h6 + \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^3\,\beta 22bar\,s3 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s2 + \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^3(\beta 22bar)^2 s2 + \left(\frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}} + \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{3}{2}} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\gamma 12bar - \frac{n2}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}} - \frac{n2\,Q2}{3\,\eta\,Q1^2}\beta 22bar\,\gamma 12bar + \frac{n2\,Q3}{3\,\eta\,Q1^2}(\beta 22bar)^{\frac{3}{2}}\gamma 12bar\right)s1 - \left(\frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}}\gamma 12bar + \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}} - \frac{n3\,Q3}{3\,\eta\,Q1^2}(\gamma 12bar)^2 - \frac{n2}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}}(\beta 22bar)^{\frac{3}{2}} - \frac{n2\,Q2}{3\,\eta\,Q1^2}(\gamma 12bar)^2\,\beta 22bar + \frac{n2\,Q3}{3\,\eta\,Q1^2}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^2\right)s3 + \frac{n2}{3\,\eta}\frac{(\beta 12)^3}{Q2}s2 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s2 + \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}}\gamma 12bar\,s3 + \frac{n3}{3\,\eta\,Q3}(\gamma 12bar)^3 s3 - \frac{n2}{3\,\eta\,Q3}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^3 s3 - \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}}s3 + \frac{n2}{3\,\eta\,Q1}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^{\frac{5}{2}}s3 + \frac{n1}{3\,\eta\,Q1}s1;$$

(*K11 and K111 should be equal *)

$$T1 = c\left(\frac{4}{7}\left(alfa1 * g1\left(\beta 12^{\frac{5}{2}} - \gamma 12bar^{\frac{5}{2}} * \beta 22bar^{\frac{5}{2}}\right) + alfa2 * g2\right)\sqrt{x2}\ (u2)^{\frac{5}{2}} +$$

$$\frac{4}{7}\,\text{alfa2}*\text{g3}\sqrt{\text{x3}}\,(\text{u3})^{\frac{5}{2}}+\frac{4}{7}\,\text{alfa1}*\text{g1}\sqrt{\text{x1}}\,(\text{u1})^{\frac{5}{2}}+$$

$$\frac{5\,\text{n2}}{21\,\eta}\left(\beta 22\text{bar}^{2}\,\frac{\text{Q2}}{\text{Q3}}\left(\text{alfa1}*\text{g1}*\gamma 12\text{bar}^{\frac{5}{2}}+\text{alfa2}*\text{g3}\right)-\right.$$

$$\left.\text{alfa1}*\text{g1}*\beta 12^{\frac{5}{2}}-\text{alfa2}*\text{g3}*\beta 22\text{bar}^{\frac{5}{2}}-\text{alfa2}*\text{g2}\right)\text{s2}+$$

$$\frac{5}{21\,\eta}\left(\frac{\gamma 12\text{bar}^{2}*\text{alfa1}*\text{g1}}{\text{Q1}}\left(\text{n3}*\text{Q3}+\beta 22\text{bar}*\text{n2}*\text{Q2}-\beta 22\text{bar}^{1.5}*\text{n2}*\text{Q3}\right)-\right.$$

$$\left.\left(\text{alfa1}*\text{g1}*\gamma 12\text{bar}^{\frac{5}{2}}+\text{alfa2}*\text{g3}\right)\left(\text{n3}+\beta 22\text{bar}\,\frac{\text{n2}*\text{Q2}}{\text{Q3}}-\beta 22\text{bar}^{1.5}*\text{n2}\right)\right)\text{s3}-$$

$$\frac{5\,\text{alfa1}\,\text{g1}}{21\,\eta}\left(\text{n1}+\gamma 12\text{bar}\,\frac{\text{n3}\,\text{Q3}}{\text{Q1}}+\gamma 12\text{bar}^{1.5}\left(\beta 22\text{bar}^{1.5}*\text{n2}-\text{n3}\right)-\right.$$

$$\left.\beta 12^{1.5}*\text{n2}-\gamma 12\text{bar}*\beta 22\text{bar}^{1.5}\,\frac{\text{n2}\,\text{Q3}}{\text{Q1}}+\gamma 12\text{bar}\,\beta 22\text{bar}\,\frac{\text{n2}\,\text{Q2}}{\text{Q1}}\right)\text{s1}\Bigg);$$

$$\text{V1}=-\text{d}\left(\frac{(\text{g1}\,\text{n1}-1)}{\text{n1}}\,\text{g1}\,\text{K11}+\frac{(\text{g3}\,\text{n3}-1)}{\text{n3}}\,\text{g3}\,\text{K33}+\right.$$

$$\left.\frac{(\text{g2}\,\text{n2}-1)}{\text{n2}}\,\text{g2}\,\text{K22}+2\,\text{g1}\,\text{g3}\,\text{K31}+\frac{4}{3}\,\text{g2}\,\text{g1}\,\text{K21}+\frac{4}{3}\,\text{g2}\,\text{g3}\,\text{K32}\right);$$

```
(*These T1 and V1 are obtained
 by integrating manually which is being verified.*)
Energy6bare = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B + n1 * g1 * 306 + n3 * g3 * 1553 + n2 * g2 * 1553;
Energy = Energy6bare;
Volumeenergy = 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Energywithoutmass = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Print["R=  ", R];
Print["normx2 (computationally)=  ", normmx2];
Print["normx2 (by hand)=  ", normx2];
Print["normx3 (computationally)=  ", normmx3];
Print["normx3 (by hand)=  ", normx3];
Print["normx1 (computationally)=  ", normmx1];
Print["normx1 (by hand)=  ", normx1];
Print["0 to x2 (h4 computationally)=  ", h4];
Print["0 to x2 (h4 by hand)=  ", AB];
```

```
 Print["x2 to x3 (h5 computationally)=  ", h5];
 Print["x2 to x3 (h5 by hand)=  ", CD];
 Print["x2 to x3 (h5 by handdddd)=  ", CCD];
 Print["x3 to x1 (h6 computationally)=  ", h6];
 Print["x3 to x1 (h6 by hand)=  ", EF];
 Print["K22  (computationally) = ", K222];
 Print["K22  (by hand) = ", K22];
 Print["K32  (computationally) = ", K332];
 Print["K32  (by hand) = ", K32];
 Print["K21  (computationally) = ", K221];
 Print["K21  (by hand) =", K21];
 Print["K31  (computationally) = ", K331];
 Print["K31  (by hand) = ", K31];
 Print["K33  (computationally) = ", K333];
 Print["K33  (by hand) = ", K33];
 Print["K11  (computationally) = ", K111];
 Print["K11  (by hand) = ", K11];
 Print["x2= ", x2];
 Print["x3= ", x3];
 Print["x1=  ", x1];
  Print["T by hand=", T1];
 Print["V by hand=", V1];
 Print["m2=", m2];
 Print["R=  ", R];
 Print["Energy without mass = ", Energywithoutmass];
 Print["Energy=  ", Energy];
 If[n == 1, If[nloop == 1, nloop = nloop + 1; SN = 1. * 10^(-1);
   Energy11 = Energy, If[nloop == 2, nloop = nloop + 1;
    H = 10^9 + 1;
    m2 = m2B;
    If[Energy - Energy11 < 0, SN = 1., SN = -1.]]]];
If[n < 3, m2 = m2 - SN * 10^(-n); Energy = F; H = F];
If[n == 3, Energy6bare = HX;
 Print["x1=", x1];
 Print["x3=", x3];
 Print["x2=", x2];
 Print["Kinetic Energy=", T1];
 Print["Potential Energy=", V1];
 Print["Volume Energy=", Volumeenergy];
]]];

(*Code for Plotting function *)
```

```
Plot1 = Plot[Evaluate[((y2[x] /.solution2) / x)^(3 / 2)],
  {x, .00001, x2}, (*%PlotRange→{{0,0.65},{0,79.5}},*)
  (*Solution 1 = f3(x3), solution3 = f3(x3), solution3 = f1(x1) *)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17]
    , StyleForm["(f2(x)/ x)^3/2", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot2 = Plot[Evaluate[((β22bar * y2[x] /.solution2) / x)^(3 / 2)], {x, .00001, x2},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f2 bar(x)/ x)^3/2", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot3 = Plot[Evaluate[((y3[x] /.solution3) / x)^(3 / 2)], {x, x2, x3},
  (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f2 bar(x)/ x)^413/2", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot4 = Plot[Evaluate[((β12 * y2[x] /.solution2) / x)^(3 / 2)], {x, .00001, x2},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^3/2", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot5 = Plot[Evaluate[((γ12bar * y3[x] /. solution3) / x)^(3 / 2)],
  {x, x2, x3}, (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^3/2", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot6 = Plot[Evaluate[((y1[x] /. solution1) / x)^(3 / 2)], {x, x3, x1},
  (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^3/2", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Show[{Plot2, Plot3}]
Show[{Plot4, Plot5, Plot6}]
```

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

General: Further output of FindMinimum::lstol will be suppressed during this calculation.

```
R=  0.0291099304
normx2 (computationally)=  -0.664963445
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678184
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305564
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.2880452
0 to x2 (h4 by hand)=  10.2880448
x2 to x3 (h5 computationally)=  2.06153971
x2 to x3 (h5 by hand)=  2.0615414
x2 to x3 (h5 by handdddd)=  2.06154059
x3 to x1 (h6 computationally)=  2.62554892
x3 to x1 (h6 by hand)=  2.62554944
K22  (computationally) = 0.74456686
K22  (by hand) = 0.744566646
K32  (computationally) = 0.70479349
K32  (by hand) = 0.704793329
K21  (computationally) = 0.454630266
K21  (by hand) =0.454630245
K31  (computationally) = 0.45054085
K31  (by hand) = 0.450541278
K33  (computationally) = 0.674379711
K33  (by hand) = 0.674381355
K11  (computationally) = 0.393038392
K11  (by hand) = 0.393039501
x2= 0.180029239
x3= 0.198998126
x1=  0.333137187
T by hand=633.496832
V by hand=-152.650345
m2=32
R=  0.0291099304
Energy without mass = 992.917908
Energy=  21112.9179
R=  0.0291099304
```

```
normx2 (computationally)=  -0.664963415
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678184
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305573
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.2913193
0 to x2 (h4 by hand)=  10.2913186
x2 to x3 (h5 computationally)=  2.06220737
x2 to x3 (h5 by hand)=  2.06220946
x2 to x3 (h5 by handdddd)=  2.06220825
x3 to x1 (h6 computationally)=  2.62641945
x3 to x1 (h6 by hand)=  2.62642007
K22  (computationally) = 0.744684475
K22  (by hand) = 0.744684118
K32  (computationally) = 0.704905095
K32  (by hand) = 0.704904828
K21  (computationally) = 0.454703703
K21  (by hand) =0.454703668
K31  (computationally) = 0.450613812
K31  (by hand) = 0.450614335
K33  (computationally) = 0.674487288
K33  (by hand) = 0.674489267
K11  (computationally) = 0.393102721
K11  (by hand) = 0.393104037
x2= 0.180000583
x3= 0.198966323
x1=  0.333082389
T by hand=633.70392
V by hand=-152.675068
m2=32.01
R=  0.0291099304
Energy without mass = 992.84762
Energy=  21112.8476
R=  0.0291099304
normx2 (computationally)=  -0.664963474
```

```
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.62867818
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305574
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.3207918
0 to x2 (h4 by hand)=  10.3207917
x2 to x3 (h5 computationally)=  2.06820943
x2 to x3 (h5 by hand)=  2.06821067
x2 to x3 (h5 by handdddd)=  2.06821031
x3 to x1 (h6 computationally)=  2.63425327
x3 to x1 (h6 by hand)=  2.63425392
K22  (computationally) = 0.745744992
K22  (by hand) = 0.745744917
K32  (computationally) = 0.705910367
K32  (by hand) = 0.70591031
K21  (computationally) = 0.455364158
K21  (by hand) =0.45536415
K31  (computationally) = 0.451268282
K31  (by hand) = 0.451268602
K33  (computationally) = 0.675449104
K33  (by hand) = 0.675450355
K11  (computationally) = 0.393678361
K11  (by hand) = 0.39367968
x2= 0.179743319
x3= 0.198680743
x1=  0.332590341
T by hand=635.56732
V by hand=-152.896546
m2=32.1
R=  0.0291099304
Energy without mass = 992.224631
Energy=  21112.2246
R=  0.0291099304
normx2 (computationally)=  -0.664963493
normx2 (by hand)=  -0.664963533
```

```
normx3 (computationally)=  -0.628678177
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305562
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.3535392
0 to x2 (h4 by hand)=  10.3535393
x2 to x3 (h5 computationally)=  2.07487972
x2 to x3 (h5 by hand)=  2.07488066
x2 to x3 (h5 by handdddd)=  2.07488061
x3 to x1 (h6 computationally)=  2.642958
x3 to x1 (h6 by hand)=  2.64295854
K22  (computationally) = 0.746921207
K22  (by hand) = 0.746921224
K32  (computationally) = 0.707025459
K32  (by hand) = 0.707025471
K21  (computationally) = 0.456096857
K21  (by hand) =0.456096859
K31  (computationally) = 0.451994586
K31  (by hand) = 0.451994832
K33  (computationally) = 0.676517015
K33  (by hand) = 0.676517996
K11  (computationally) = 0.394316631
K11  (by hand) = 0.394317719
x2= 0.179458748
x3= 0.198364865
x1=  0.332046127
T by hand=637.637859
V by hand=-153.142216
m2=32.2
R=  0.0291099304
Energy without mass = 991.552254
Energy=  21111.5523
R=  0.0291099304
normx2 (computationally)=  -0.664963494
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678174
```

```
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305562
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.3862873
0 to x2 (h4 by hand)=  10.3862873
x2 to x3 (h5 computationally)=  2.08155078
x2 to x3 (h5 by hand)=  2.08155167
x2 to x3 (h5 by handdddd)=  2.08155167
x3 to x1 (h6 computationally)=  2.65166367
x3 to x1 (h6 by hand)=  2.65166423
K22  (computationally) = 0.748095433
K22  (by hand) = 0.748095452
K32  (computationally) = 0.708138727
K32  (by hand) = 0.708138741
K21  (computationally) = 0.456828429
K21  (by hand) =0.456828431
K31  (computationally) = 0.452719875
K31  (by hand) = 0.452720109
K33  (computationally) = 0.677583611
K33  (by hand) = 0.677584548
K11  (computationally) = 0.394954314
K11  (by hand) = 0.39495543
x2= 0.179175515
x3= 0.198050479
x1=  0.331504543
T by hand=639.70859
V by hand=-153.387607
m2=32.3
R=  0.0291099304
Energy without mass = 990.900536
Energy=  21110.9005
```

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

```
R=  0.0291099304
normx2 (computationally)=  -0.66496346
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678173
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305562
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.4190356
0 to x2 (h4 by hand)=  10.4190354
x2 to x3 (h5 computationally)=  2.08822296
x2 to x3 (h5 by hand)=  2.08822429
x2 to x3 (h5 by handdddd)=  2.08822385
x3 to x1 (h6 computationally)=  2.66037009
x3 to x1 (h6 by hand)=  2.66037065
K22  (computationally) = 0.749267548
K22  (by hand) = 0.749267404
K32  (computationally) = 0.709250106
K32  (by hand) = 0.709249998
K21  (computationally) = 0.457558846
K21  (by hand) =0.457558832
K31  (computationally) = 0.453444207
K31  (by hand) = 0.453444547
K33  (computationally) = 0.678649183
K33  (by hand) = 0.678650493
K11  (computationally) = 0.395591037
K11  (by hand) = 0.395592203
x2= 0.178893607
x3= 0.197737577
x1=  0.330965563
T by hand=641.779471
V by hand=-153.63265
m2=32.4
R=  0.0291099304
Energy without mass = 990.269245
Energy=  21110.2692
```

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

General: Further output of FindMinimum::fmdig will be suppressed during this calculation.

```
R=  0.0291099304
normx2 (computationally)=  -0.664963489
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678173
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305563
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.4517859
0 to x2 (h4 by hand)=  10.4517859
x2 to x3 (h5 computationally)=  2.09489414
x2 to x3 (h5 by hand)=  2.09489508
x2 to x3 (h5 by handdddd)=  2.09489504
x3 to x1 (h6 computationally)=  2.66907725
x3 to x1 (h6 by hand)=  2.66907783
K22  (computationally) = 0.750438401
K22  (by hand) = 0.750438399
K32  (computationally) = 0.710360085
K32  (by hand) = 0.710360084
K21  (computationally) = 0.458288169
K21  (by hand) =0.458288169
K31  (computationally) = 0.45416713
K31  (by hand) = 0.454167376
K33  (computationally) = 0.679712051
K33  (by hand) = 0.679713026
K11  (computationally) = 0.396226714
K11  (by hand) = 0.396227883
x2= 0.178613024
x3= 0.197426137
x1=  0.330429162
T by hand=643.850464
V by hand=-153.877259
m2=32.5
R=  0.0291099304
```

```
Energy without mass = 989.658194
Energy=  21109.6582
R=  0.0291099304
normx2 (computationally)=  -0.664963493
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678173
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305563
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.4845367
0 to x2 (h4 by hand)=  10.4845367
x2 to x3 (h5 computationally)=  2.10156622
x2 to x3 (h5 by hand)=  2.10156711
x2 to x3 (h5 by handdddd)=  2.10156713
x3 to x1 (h6 computationally)=  2.67778511
x3 to x1 (h6 by hand)=  2.67778569
K22  (computationally) = 0.751607232
K22  (by hand) = 0.751607249
K32  (computationally) = 0.711468231
K32  (by hand) = 0.711468243
K21  (computationally) = 0.459016351
K21  (by hand) =0.459016352
K31  (computationally) = 0.454889056
K31  (by hand) = 0.45488929
K33  (computationally) = 0.680773727
K33  (by hand) = 0.680774662
K11  (computationally) = 0.396861387
K11  (by hand) = 0.396862548
x2= 0.178333747
x3= 0.197116157
x1=  0.329895323
T by hand=645.921595
V by hand=-154.121503
m2=32.6
R=  0.0291099304
Energy without mass = 989.067127
```

```
Energy=  21109.0671
R=  0.0291099304
normx2 (computationally)=  -0.664963451
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678172
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305562
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.5172877
0 to x2 (h4 by hand)=  10.5172873
x2 to x3 (h5 computationally)=  2.10823962
x2 to x3 (h5 by hand)=  2.10824108
x2 to x3 (h5 by handdddd)=  2.10824053
x3 to x1 (h6 computationally)=  2.68649371
x3 to x1 (h6 by hand)=  2.68649428
K22  (computationally) = 0.752773878
K22  (by hand) = 0.752773694
K32  (computationally) = 0.712574452
K32  (by hand) = 0.712574314
K21  (computationally) = 0.459743387
K21  (by hand) =0.459743369
K31  (computationally) = 0.455610086
K31  (by hand) = 0.455610456
K33  (computationally) = 0.681834563
K33  (by hand) = 0.681835983
K11  (computationally) = 0.397495121
K11  (by hand) = 0.397496303
x2= 0.178055765
x3= 0.196807626
x1=  0.329364026
T by hand=647.992875
V by hand=-154.365412
m2=32.7
R=  0.0291099304
Energy without mass = 988.495818
Energy=  21108.4958
```

```
R=  0.0291099304
normx2 (computationally)=  -0.664963493
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678172
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305563
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.5500408
0 to x2 (h4 by hand)=  10.5500408
x2 to x3 (h5 computationally)=  2.11491156
x2 to x3 (h5 by hand)=  2.11491245
x2 to x3 (h5 by handdddd)=  2.11491247
x3 to x1 (h6 computationally)=  2.69520306
x3 to x1 (h6 by hand)=  2.69520364
K22  (computationally) = 0.753939459
K22  (by hand) = 0.753939477
K32  (computationally) = 0.713679393
K32  (by hand) = 0.713679406
K21  (computationally) = 0.460469359
K21  (by hand) =0.46046936
K31  (computationally) = 0.456329621
K31  (by hand) = 0.456329858
K33  (computationally) = 0.682892334
K33  (by hand) = 0.68289328
K11  (computationally) = 0.398127846
K11  (by hand) = 0.398129017
x2= 0.177779078
x3= 0.196500521
x1=  0.328835249
T by hand=650.064264
V by hand=-154.608881
m2=32.8
R=  0.0291099304
Energy without mass = 987.944102
Energy=  21107.9441
R=  0.0291099304
```

normx2 (computationally)= -0.664963475

normx2 (by hand)= -0.664963533

normx3 (computationally)= -0.628678172

normx3 (by hand)= -0.628678207

normx1 (computationally)= -0.397305573

normx1 (by hand)= -0.397305574

0 to x2 (h4 computationally)= 10.5827939

0 to x2 (h4 by hand)= 10.5827938

x2 to x3 (h5 computationally)= 2.12158511

x2 to x3 (h5 by hand)= 2.12158625

x2 to x3 (h5 by handdddd)= 2.12158603

x3 to x1 (h6 computationally)= 2.70391332

x3 to x1 (h6 by hand)= 2.70391403

K22 (computationally) = 0.755102752

K22 (by hand) = 0.755102684

K32 (computationally) = 0.714782357

K32 (by hand) = 0.714782306

K21 (computationally) = 0.461194208

K21 (by hand) =0.461194201

K31 (computationally) = 0.457048357

K31 (by hand) = 0.457048652

K33 (computationally) = 0.683949521

K33 (by hand) = 0.683950676

K11 (computationally) = 0.398760009

K11 (by hand) = 0.398761431

x2= 0.177503665

x3= 0.196194845

x1= 0.32830898

T by hand=652.135851

V by hand=-154.852109

m2=32.9

R= 0.0291099304

Energy without mass = 987.411706

Energy= 21107.4117

R= 0.0291099304

normx2 (computationally)= -0.66496344

```
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678172
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305562
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.6155476
0 to x2 (h4 by hand)=  10.6155471
x2 to x3 (h5 computationally)=  2.12825938
x2 to x3 (h5 by hand)=  2.12826102
x2 to x3 (h5 by handdddd)=  2.12826031
x3 to x1 (h6 computationally)=  2.7126239
x3 to x1 (h6 by hand)=  2.71262448
K22  (computationally) = 0.756264123
K22  (by hand) = 0.756263884
K32  (computationally) = 0.715883556
K32  (by hand) = 0.715883377
K21  (computationally) = 0.461917899
K21  (by hand) =0.461917875
K31  (computationally) = 0.457766038
K31  (by hand) = 0.45776645
K33  (computationally) = 0.685005412
K33  (by hand) = 0.685006982
K11  (computationally) = 0.39939047
K11  (by hand) = 0.399391678
x2= 0.177229518
x3= 0.195890583
x1=  0.327785181
T by hand=654.20748
V by hand=-155.094816
m2=33.
R=  0.0291099304
Energy without mass = 986.898449
Energy=  21106.8984
R=  0.0291099304
normx2 (computationally)=  -0.664963493
normx2 (by hand)=  -0.664963533
```

```
normx3 (computationally)=  -0.628678172
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305573
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.6483036
0 to x2 (h4 by hand)=  10.6483036
x2 to x3 (h5 computationally)=  2.13493207
x2 to x3 (h5 by hand)=  2.13493298
x2 to x3 (h5 by handdddd)=  2.134933
x3 to x1 (h6 computationally)=  2.72133562
x3 to x1 (h6 by hand)=  2.72133633
K22  (computationally) = 0.757424501
K22  (by hand) = 0.757424519
K32  (computationally) = 0.716983527
K32  (by hand) = 0.71698354
K21  (computationally) = 0.462640586
K21  (by hand) =0.462640588
K31  (computationally) = 0.458482256
K31  (by hand) = 0.458482497
K33  (computationally) = 0.686058164
K33  (by hand) = 0.686059125
K11  (computationally) = 0.400020657
K11  (by hand) = 0.400022075
x2= 0.176956639
x3= 0.195587714
x1=  0.327263855
T by hand=656.279307
V by hand=-155.337244
m2=33.1
R=  0.0291099304
Energy without mass = 986.40415
Energy=  21106.4041
R=  0.0291099304
normx2 (computationally)=  -0.664963439
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678172
```

```
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305574
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.6810587
0 to x2 (h4 by hand)=  10.6810582
x2 to x3 (h5 computationally)=  2.14160742
x2 to x3 (h5 by hand)=  2.14160908
x2 to x3 (h5 by handdddd)=  2.14160836
x3 to x1 (h6 computationally)=  2.73004785
x3 to x1 (h6 by hand)=  2.73004857
K22  (computationally) = 0.758582197
K22  (by hand) = 0.758581951
K32  (computationally) = 0.718081302
K32  (by hand) = 0.718081118
K21  (computationally) = 0.463362097
K21  (by hand) =0.463362073
K31  (computationally) = 0.459197877
K31  (by hand) = 0.459198295
K33  (computationally) = 0.687111194
K33  (by hand) = 0.68711279
K11  (computationally) = 0.400649635
K11  (by hand) = 0.400651115
x2= 0.176684999
x3= 0.195286246
x1=  0.326744969
T by hand=658.351258
V by hand=-155.579324
m2=33.2
R=  0.0291099304
Energy without mass = 985.92854
Energy=  21105.9285
R=  0.0291099304
normx2 (computationally)=  -0.664963424
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678172
normx3 (by hand)=  -0.628678207
```

```
normx1 (computationally)=  -0.397305571
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.7138154
0 to x2 (h4 by hand)=  10.7138147
x2 to x3 (h5 computationally)=  2.14828226
x2 to x3 (h5 by hand)=  2.14828414
x2 to x3 (h5 by handdddd)=  2.1482832
x3 to x1 (h6 computationally)=  2.73876074
x3 to x1 (h6 by hand)=  2.73876143
K22  (computationally) = 0.759738487
K22  (by hand) = 0.759738167
K32  (computationally) = 0.719177619
K32  (by hand) = 0.719177379
K21  (computationally) = 0.464082535
K21  (by hand) =0.464082503
K31  (computationally) = 0.459912228
K31  (by hand) = 0.459912699
K33  (computationally) = 0.688161967
K33  (by hand) = 0.688163751
K11  (computationally) = 0.401277509
K11  (by hand) = 0.401278947
x2= 0.176414603
x3= 0.194986155
x1=  0.326228504
T by hand=660.423309
V by hand=-155.820978
m2=33.3
R=  0.0291099304
Energy without mass = 985.471474
Energy=  21105.4715
R=  0.0291099304
normx2 (computationally)=  -0.664963426
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678172
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305565
```

```
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.7465731
0 to x2 (h4 by hand)=  10.7465725
x2 to x3 (h5 computationally)=  2.15495705
x2 to x3 (h5 by hand)=  2.15495891
x2 to x3 (h5 by handdddd)=  2.154958
x3 to x1 (h6 computationally)=  2.74747428
x3 to x1 (h6 by hand)=  2.7474749
K22  (computationally) = 0.760893186
K22  (by hand) = 0.760892873
K32  (computationally) = 0.720272371
K32  (by hand) = 0.720272137
K21  (computationally) = 0.464801891
K21  (by hand) =0.46480186
K31  (computationally) = 0.460625415
K31  (by hand) = 0.460625881
K33  (computationally) = 0.689210858
K33  (by hand) = 0.689212626
K11  (computationally) = 0.4019043
K11  (by hand) = 0.401905609
x2= 0.176145439
x3= 0.194687431
x1=  0.325714443
T by hand=662.495469
V by hand=-156.062225
m2=33.4
R=  0.0291099304
Energy without mass = 985.032746
Energy=  21105.0327
R=  0.0291099304
normx2 (computationally)=  -0.664963414
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678187
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305562
normx1 (by hand)=  -0.397305574
```

```
0 to x2 (h4 computationally)=  10.7793315
0 to x2 (h4 by hand)=  10.7793308
x2 to x3 (h5 computationally)=  2.16163288
x2 to x3 (h5 by hand)=  2.16163517
x2 to x3 (h5 by handdddd)=  2.16163383
x3 to x1 (h6 computationally)=  2.75618847
x3 to x1 (h6 by hand)=  2.756189
K22  (computationally) = 0.762046037
K22  (by hand) = 0.762045664
K32  (computationally) = 0.721365444
K32  (by hand) = 0.721365164
K21  (computationally) = 0.46552016
K21  (by hand) =0.465520123
K31  (computationally) = 0.461337696
K31  (by hand) = 0.46133827
K33  (computationally) = 0.690258847
K33  (by hand) = 0.690261027
K11  (computationally) = 0.402530117
K11  (by hand) = 0.402531263
x2= 0.175877495
x3= 0.194390072
x1=  0.325202768
T by hand=664.567767
V by hand=-156.303136
m2=33.5
R=  0.0291099304
Energy without mass = 984.612138
Energy=  21104.6121
R=  0.0291099304
normx2 (computationally)=  -0.664963493
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678184
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305569
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.8120922
```

```
0 to x2 (h4 by hand)=  10.8120923
x2 to x3 (h5 computationally)=  2.16830651
x2 to x3 (h5 by hand)=  2.16830764
x2 to x3 (h5 by handdddd)=  2.16830747
x3 to x1 (h6 computationally)=  2.76490368
x3 to x1 (h6 by hand)=  2.76490429
K22  (computationally) = 0.763197981
K22  (by hand) = 0.763198005
K32  (computationally) = 0.722457321
K32  (by hand) = 0.722457339
K21  (computationally) = 0.466237434
K21  (by hand) =0.466237436
K31  (computationally) = 0.462048377
K31  (by hand) = 0.462048675
K33  (computationally) = 0.691303138
K33  (by hand) = 0.69130433
K11  (computationally) = 0.403155402
K11  (by hand) = 0.403156649
x2= 0.175610771
x3= 0.194094048
x1=  0.324693468
T by hand=666.640211
V by hand=-156.543685
m2=33.6
R=  0.0291099304
Energy without mass = 984.209511
Energy=  21104.2095
R=  0.0291099304
normx2 (computationally)=  -0.664963425
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678182
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305566
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.8448515
0 to x2 (h4 by hand)=  10.8448508
```

```
x2 to x3 (h5 computationally)=  2.17498378
x2 to x3 (h5 by hand)=  2.17498585
x2 to x3 (h5 by handdddd)=  2.17498474
x3 to x1 (h6 computationally)=  2.7736194
x3 to x1 (h6 by hand)=  2.77362
K22  (computationally) = 0.764346898
K22  (by hand) = 0.76434658
K32  (computationally) = 0.723546819
K32  (by hand) = 0.72354658
K21  (computationally) = 0.466953534
K21  (by hand) =0.466953503
K31  (computationally) = 0.462758704
K31  (by hand) = 0.462759222
K33  (computationally) = 0.692348532
K33  (by hand) = 0.692350506
K11  (computationally) = 0.403779604
K11  (by hand) = 0.403780872
x2= 0.175345237
x3= 0.193799379
x1=  0.324186518
T by hand=668.7128
V by hand=-156.783947
m2=33.7
R=  0.0291099304
Energy without mass = 983.824579
Energy=  21103.8246
R=  0.0291099304
normx2 (computationally)=  -0.664963463
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.62867818
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305573
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.8776132
0 to x2 (h4 by hand)=  10.877613
x2 to x3 (h5 computationally)=  2.181659
```

```
x2 to x3 (h5 by hand)=  2.18166049
x2 to x3 (h5 by handdddd)=  2.18165996
x3 to x1 (h6 computationally)=  2.78233601
x3 to x1 (h6 by hand)=  2.7823367
K22  (computationally) = 0.765495061
K22  (by hand) = 0.765494934
K32  (computationally) = 0.724635248
K32  (by hand) = 0.724635153
K21  (computationally) = 0.467668651
K21  (by hand) =0.467668638
K31  (computationally) = 0.463467473
K31  (by hand) = 0.463467856
K33  (computationally) = 0.69339047
K33  (by hand) = 0.693391957
K11  (computationally) = 0.404403085
K11  (by hand) = 0.404404496
x2= 0.175080907
x3= 0.193506027
x1=  0.323681903
T by hand=670.785517
V by hand=-157.023823
m2=33.8
R=  0.0291099304
Energy without mass = 983.457249
Energy=  21103.4572
R=  0.0291099304
normx2 (computationally)=  -0.664963477
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678178
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305573
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.9103755
0 to x2 (h4 by hand)=  10.9103754
x2 to x3 (h5 computationally)=  2.18833509
x2 to x3 (h5 by hand)=  2.18833636
```

x2 to x3 (h5 by handdddd)= 2.18833606

x3 to x1 (h6 computationally)= 2.79105319

x3 to x1 (h6 by hand)= 2.7910539

K22 (computationally) = 0.766641308

K22 (by hand) = 0.766641252

K32 (computationally) = 0.725721942

K32 (by hand) = 0.7257219

K21 (computationally) = 0.46838268

K21 (by hand) =0.468382675

K31 (computationally) = 0.464175301

K31 (by hand) = 0.464175631

K33 (computationally) = 0.694431313

K33 (by hand) = 0.694432608

K11 (computationally) = 0.405025452

K11 (by hand) = 0.405026887

x2= 0.174817759

x3= 0.193213997

x1= 0.3231796

T by hand=672.858343

V by hand=−157.263319

m2=33.9

R= 0.0291099304

Energy without mass = 983.10729

Energy= 21103.1073

R= 0.0291099304

normx2 (computationally)= −0.664963446

normx2 (by hand)= −0.664963533

normx3 (computationally)= −0.628678176

normx3 (by hand)= −0.628678207

normx1 (computationally)= −0.397305574

normx1 (by hand)= −0.397305574

0 to x2 (h4 computationally)= 10.9431378

0 to x2 (h4 by hand)= 10.9431374

x2 to x3 (h5 computationally)= 2.19501252

x2 to x3 (h5 by hand)= 2.19501422

x2 to x3 (h5 by handdddd)= 2.19501349

```
x3 to x1 (h6 computationally)=  2.79977106
x3 to x1 (h6 by hand)=  2.79977178
K22  (computationally) = 0.767785459
K22  (by hand) = 0.767785245
K32  (computationally) = 0.726806801
K32  (by hand) = 0.72680664
K21  (computationally) = 0.469095628
K21  (by hand) =0.469095607
K31  (computationally) = 0.464882307
K31  (by hand) = 0.464882738
K33  (computationally) = 0.695471445
K33  (by hand) = 0.695473098
K11  (computationally) = 0.405646947
K11  (by hand) = 0.405648441
x2= 0.174555784
x3= 0.192923281
x1=  0.322679596
T by hand=674.931312
V by hand=-157.502505
m2=34.
R=  0.0291099304
Energy without mass = 982.774508
Energy=  21102.7745
R=  0.0291099304
normx2 (computationally)=  -0.664963493
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678175
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305561
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  10.9759022
0 to x2 (h4 by hand)=  10.9759023
x2 to x3 (h5 computationally)=  2.2016885
x2 to x3 (h5 by hand)=  2.20168951
x2 to x3 (h5 by handdddd)=  2.20168949
x3 to x1 (h6 computationally)=  2.80848937
```

```
x3 to x1 (h6 by hand)=  2.80848996
K22  (computationally) = 0.768928634
K22  (by hand) = 0.768928658
K32  (computationally) = 0.727890471
K32  (by hand) = 0.727890488
K21  (computationally) = 0.469807545
K21  (by hand) =0.469807547
K31  (computationally) = 0.465587859
K31  (by hand) = 0.465588126
K33  (computationally) = 0.696508631
K33  (by hand) = 0.696509698
K11  (computationally) = 0.406266967
K11  (by hand) = 0.406268159
x2= 0.174294983
x3= 0.192633857
x1=  0.322181861
T by hand=677.004316
V by hand=-157.741159
m2=34.1
R=  0.0291099304
Energy without mass = 982.458785
Energy=  21102.4588
R=  0.0291099304
normx2 (computationally)=  -0.664963492
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678174
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305568
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.0086668
0 to x2 (h4 by hand)=  11.0086668
x2 to x3 (h5 computationally)=  2.20836588
x2 to x3 (h5 by hand)=  2.20836689
x2 to x3 (h5 by handdddd)=  2.20836687
x3 to x1 (h6 computationally)=  2.81720872
x3 to x1 (h6 by hand)=  2.81720939
```

```
K22  (computationally) = 0.770069704
K22  (by hand) = 0.770069726
K32  (computationally) = 0.728972306
K32  (by hand) = 0.728972322
K21  (computationally) = 0.470518425
K21  (by hand) =0.470518427
K31  (computationally) = 0.466292646
K31  (by hand) = 0.466292912
K33  (computationally) = 0.697545163
K33  (by hand) = 0.697546222
K11  (computationally) = 0.406886787
K11  (by hand) = 0.406888133
x2= 0.174035337
x3= 0.192345726
x1=  0.321686402
T by hand=679.077548
V by hand=-157.979654
m2=34.2
R=  0.0291099304
Energy without mass = 982.159868
Energy=  21102.1599
R=  0.0291099304
normx2 (computationally)=  -0.664963462
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678173
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305563
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.0414316
0 to x2 (h4 by hand)=  11.0414313
x2 to x3 (h5 computationally)=  2.21504426
x2 to x3 (h5 by hand)=  2.21504569
x2 to x3 (h5 by handdddd)=  2.21504525
x3 to x1 (h6 computationally)=  2.82592853
x3 to x1 (h6 by hand)=  2.82592915
K22  (computationally) = 0.771208838
```

```
K22  (by hand) = 0.771208705
K32  (computationally) = 0.730052404
K32  (by hand) = 0.730052304
K21  (computationally) = 0.471228224
K21  (by hand) =0.47122821
K31  (computationally) = 0.466996528
K31  (by hand) = 0.466996894
K33  (computationally) = 0.698580727
K33  (by hand) = 0.698582143
K11  (computationally) = 0.407505338
K11  (by hand) = 0.407506614
x2= 0.173776839
x3= 0.19205888
x1=  0.321193184
T by hand=681.150867
V by hand=-158.217742
m2=34.3
R=  0.0291099304
Energy without mass = 981.87761
Energy=  21101.8776
R=  0.0291099304
normx2 (computationally)=  -0.664963492
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678172
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305565
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.0741983
0 to x2 (h4 by hand)=  11.0741983
x2 to x3 (h5 computationally)=  2.22172159
x2 to x3 (h5 by hand)=  2.22172257
x2 to x3 (h5 by handdddd)=  2.22172258
x3 to x1 (h6 computationally)=  2.83464916
x3 to x1 (h6 by hand)=  2.83464981
K22  (computationally) = 0.772346856
K22  (by hand) = 0.772346877
```

```
K32  (computationally) = 0.731131242
K32  (by hand) = 0.731131258
K21  (computationally) = 0.471937034
K21  (by hand) =0.471937036
K31  (computationally) = 0.467699096
K31  (by hand) = 0.467699357
K33  (computationally) = 0.699613685
K33  (by hand) = 0.699614727
K11  (computationally) = 0.40812315
K11  (by hand) = 0.408124452
x2= 0.173519488
x3= 0.191773298
x1=  0.320702197
T by hand=683.224315
V by hand=-158.455475
m2=34.4
R=  0.0291099304
Energy without mass = 981.611864
Energy=  21101.6119
R=  0.0291099304
normx2 (computationally)=  -0.664963459
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305564
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.1069647
0 to x2 (h4 by hand)=  11.1069644
x2 to x3 (h5 computationally)=  2.22840067
x2 to x3 (h5 by hand)=  2.22840213
x2 to x3 (h5 by handdddd)=  2.22840167
x3 to x1 (h6 computationally)=  2.84337042
x3 to x1 (h6 by hand)=  2.84337107
K22  (computationally) = 0.773482648
K22  (by hand) = 0.773482499
K32  (computationally) = 0.732208183
```

K32 (by hand) = 0.732208071

K21 (computationally) = 0.472644771

K21 (by hand) =0.472644756

K31 (computationally) = 0.468400954

K31 (by hand) = 0.468401327

K33 (computationally) = 0.700646305

K33 (by hand) = 0.700647746

K11 (computationally) = 0.408740069

K11 (by hand) = 0.408741411

x2= 0.173263265

x3= 0.191488984

x1= 0.320213423

T by hand=685.297904

V by hand=-158.69291

m2=34.5

R= 0.0291099304

Energy without mass = 981.362412

Energy= 21101.3624

R= 0.0291099304

normx2 (computationally)= -0.664963461

normx2 (by hand)= -0.664963533

normx3 (computationally)= -0.628678171

normx3 (by hand)= -0.628678207

normx1 (computationally)= -0.39730557

normx1 (by hand)= -0.397305574

0 to x2 (h4 computationally)= 11.1397326

0 to x2 (h4 by hand)= 11.1397323

x2 to x3 (h5 computationally)= 2.23507928

x2 to x3 (h5 by hand)= 2.23508072

x2 to x3 (h5 by handdddd)= 2.23508029

x3 to x1 (h6 computationally)= 2.85209245

x3 to x1 (h6 by hand)= 2.85209317

K22 (computationally) = 0.774617111

K22 (by hand) = 0.774616971

K32 (computationally) = 0.733283747

K32 (by hand) = 0.733283642

```
K21  (computationally) = 0.473351511
K21  (by hand) =0.473351497
K31  (computationally) = 0.469101632
K31  (by hand) = 0.469102
K33  (computationally) = 0.701676805
K33  (by hand) = 0.701678231
K11  (computationally) = 0.40935623
K11  (by hand) = 0.409357694
x2= 0.17300817
x3= 0.191205918
x1=  0.319726847
T by hand=687.371625
V by hand=-158.930006
m2=34.6
R=  0.0291099304
Energy without mass = 981.129124
Energy=  21101.1291
R=  0.0291099304
normx2 (computationally)=  -0.664963492
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305574
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.1725017
0 to x2 (h4 by hand)=  11.1725018
x2 to x3 (h5 computationally)=  2.24175751
x2 to x3 (h5 by hand)=  2.24175849
x2 to x3 (h5 by handdddd)=  2.24175852
x3 to x1 (h6 computationally)=  2.86081513
x3 to x1 (h6 by hand)=  2.86081591
K22  (computationally) = 0.775750212
K22  (by hand) = 0.775750237
K32  (computationally) = 0.734357917
K32  (by hand) = 0.734357936
K21  (computationally) = 0.474057244
```

K21 (by hand) =0.474057246

K31 (computationally) = 0.46980114

K31 (by hand) = 0.4698014

K33 (computationally) = 0.702705257

K33 (by hand) = 0.702706295

K11 (computationally) = 0.40997142

K11 (by hand) = 0.409972954

x2= 0.172754195

x3= 0.190924093

x1= 0.319242449

T by hand=689.445451

V by hand=−159.16672

m2=34.7

R= 0.0291099304

Energy without mass = 980.911837

Energy= 21100.9118

R= 0.0291099304

normx2 (computationally)= −0.664963447

normx2 (by hand)= −0.664963533

normx3 (computationally)= −0.628678171

normx3 (by hand)= −0.628678207

normx1 (computationally)= −0.397305574

normx1 (by hand)= −0.397305574

0 to x2 (h4 computationally)= 11.2052704

0 to x2 (h4 by hand)= 11.20527

x2 to x3 (h5 computationally)= 2.24843783

x2 to x3 (h5 by hand)= 2.24843948

x2 to x3 (h5 by handdddd)= 2.24843885

x3 to x1 (h6 computationally)= 2.8695384

x3 to x1 (h6 by hand)= 2.86953918

K22 (computationally) = 0.776880971

K22 (by hand) = 0.776880761

K32 (computationally) = 0.735430129

K32 (by hand) = 0.735429971

K21 (computationally) = 0.474761902

K21 (by hand) =0.474761881

```
K31  (computationally) = 0.470500021
K31  (by hand) = 0.47050044
K33  (computationally) = 0.703733661
K33  (by hand) = 0.703735266
K11  (computationally) = 0.410585667
K11  (by hand) = 0.410587246
x2= 0.172501323
x3= 0.19064351
x1=  0.318760215
T by hand=691.519411
V by hand=-159.403137
m2=34.8
R=  0.0291099304
Energy without mass = 980.710342
Energy=  21100.7103
R=  0.0291099304
normx2 (computationally)=  -0.66496348
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305574
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.2380412
0 to x2 (h4 by hand)=  11.2380412
x2 to x3 (h5 computationally)=  2.25511663
x2 to x3 (h5 by hand)=  2.2551178
x2 to x3 (h5 by handdddd)=  2.25511766
x3 to x1 (h6 computationally)=  2.87826235
x3 to x1 (h6 by hand)=  2.87826313
K22  (computationally) = 0.778010831
K22  (by hand) = 0.778010794
K32  (computationally) = 0.73650122
K32  (by hand) = 0.736501193
K21  (computationally) = 0.475465595
K21  (by hand) =0.475465591
K31  (computationally) = 0.471197498
```

K31 (by hand) = 0.471197801

K33 (computationally) = 0.704759123

K33 (by hand) = 0.70476032

K11 (computationally) = 0.411198952

K11 (by hand) = 0.411200516

x2= 0.172249558

x3= 0.190364144

x1= 0.318280126

T by hand=693.593463

V by hand=−159.639139

m2=34.9

R= 0.0291099304

Energy without mass = 980.524543

Energy= 21100.5245

R= 0.0291099304

normx2 (computationally)= −0.664963491

normx2 (by hand)= −0.664963533

normx3 (computationally)= −0.628678171

normx3 (by hand)= −0.628678207

normx1 (computationally)= −0.397305563

normx1 (by hand)= −0.397305574

0 to x2 (h4 computationally)= 11.2708125

0 to x2 (h4 by hand)= 11.2708125

x2 to x3 (h5 computationally)= 2.26179627

x2 to x3 (h5 by hand)= 2.26179729

x2 to x3 (h5 by handdddd)= 2.2617973

x3 to x1 (h6 computationally)= 2.88698673

x3 to x1 (h6 by hand)= 2.88698738

K22 (computationally) = 0.779138862

K22 (by hand) = 0.77913888

K32 (computationally) = 0.737570655

K32 (by hand) = 0.737570668

K21 (computationally) = 0.47616824

K21 (by hand) =0.476168242

K31 (computationally) = 0.471894065

K31 (by hand) = 0.471894334

```
K33  (computationally) = 0.705783541
K33  (by hand) = 0.705784613
K11  (computationally) = 0.411810957
K11  (by hand) = 0.411812258
x2= 0.171998884
x3= 0.190085997
x1=  0.317802163
T by hand=695.66759
V by hand=-159.874731
m2=35.
R=  0.0291099304
Energy without mass = 980.354246
Energy=  21100.3542
R=  0.0291099304
normx2 (computationally)=  -0.664963446
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305573
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.3035836
0 to x2 (h4 by hand)=  11.3035832
x2 to x3 (h5 computationally)=  2.26847751
x2 to x3 (h5 by hand)=  2.2684792
x2 to x3 (h5 by handdddd)=  2.26847855
x3 to x1 (h6 computationally)=  2.8957122
x3 to x1 (h6 by hand)=  2.89571298
K22  (computationally) = 0.780264766
K22  (by hand) = 0.780264548
K32  (computationally) = 0.738638264
K32  (by hand) = 0.7386381
K21  (computationally) = 0.47686989
K21  (by hand) =0.476869868
K31  (computationally) = 0.472589964
K31  (by hand) = 0.472590392
K33  (computationally) = 0.706807538
```

K33 (by hand) = 0.706809179

K11 (computationally) = 0.412422959

K11 (by hand) = 0.412424558

x2= 0.171749291

x3= 0.189809062

x1= 0.317326331

T by hand=697.741964

V by hand=−160.110219

m2=35.1

R= 0.0291099304

Energy without mass = 980.19925

Energy= 21100.1992

R= 0.0291099304

normx2 (computationally)= −0.664963488

normx2 (by hand)= −0.664963533

normx3 (computationally)= −0.628678171

normx3 (by hand)= −0.628678207

normx1 (computationally)= −0.397305563

normx1 (by hand)= −0.397305574

0 to x2 (h4 computationally)= 11.336357

0 to x2 (h4 by hand)= 11.336357

x2 to x3 (h5 computationally)= 2.27515702

x2 to x3 (h5 by hand)= 2.27515809

x2 to x3 (h5 by handdddd)= 2.27515806

x3 to x1 (h6 computationally)= 2.90443792

x3 to x1 (h6 by hand)= 2.90443859

K22 (computationally) = 0.781389875

K22 (by hand) = 0.781389878

K32 (computationally) = 0.73970482

K32 (by hand) = 0.739704822

K21 (computationally) = 0.477570549

K21 (by hand) =0.47757055

K31 (computationally) = 0.473284382

K31 (by hand) = 0.473284664

K33 (computationally) = 0.707828441

K33 (by hand) = 0.707829562

```
K11  (computationally) = 0.413033221
K11  (by hand) = 0.413034557
x2= 0.171500782
x3= 0.189533318
x1=  0.316852584
T by hand=699.816324
V by hand=-160.345116
m2=35.2
R=  0.0291099304
Energy without mass = 980.059501
Energy=  21100.0595
R=  0.0291099304
normx2 (computationally)=  -0.664963489
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305561
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.3691304
0 to x2 (h4 by hand)=  11.3691305
x2 to x3 (h5 computationally)=  2.28183778
x2 to x3 (h5 by hand)=  2.28183884
x2 to x3 (h5 by handdddd)=  2.28183883
x3 to x1 (h6 computationally)=  2.91316445
x3 to x1 (h6 by hand)=  2.91316509
K22  (computationally) = 0.782513009
K22  (by hand) = 0.782513019
K32  (computationally) = 0.740769643
K32  (by hand) = 0.74076965
K21  (computationally) = 0.478270203
K21  (by hand) =0.478270204
K31  (computationally) = 0.473978036
K31  (by hand) = 0.473978314
K33  (computationally) = 0.708848658
K33  (by hand) = 0.708849768
K11  (computationally) = 0.413642948
```

K11 (by hand) = 0.413644239

x2= 0.17125334

x3= 0.189258768

x1= 0.316380929

T by hand=701.890856

V by hand=−160.579782

m2=35.3

R= 0.0291099304

Energy without mass = 979.934764

Energy= 21099.9348

R= 0.0291099304

normx2 (computationally)= −0.664963468

normx2 (by hand)= −0.664963533

normx3 (computationally)= −0.628678171

normx3 (by hand)= −0.628678207

normx1 (computationally)= −0.397305561

normx1 (by hand)= −0.397305574

0 to x2 (h4 computationally)= 11.4019043

0 to x2 (h4 by hand)= 11.4019041

x2 to x3 (h5 computationally)= 2.28851936

x2 to x3 (h5 by hand)= 2.28852073

x2 to x3 (h5 by handdddd)= 2.28852041

x3 to x1 (h6 computationally)= 2.92189167

x3 to x1 (h6 by hand)= 2.92189231

K22 (computationally) = 0.783634348

K22 (by hand) = 0.783634249

K32 (computationally) = 0.741832839

K32 (by hand) = 0.741832765

K21 (computationally) = 0.478968854

K21 (by hand) =0.478968844

K31 (computationally) = 0.474670822

K31 (by hand) = 0.474671175

K33 (computationally) = 0.709867844

K33 (by hand) = 0.709869219

K11 (computationally) = 0.414251915

K11 (by hand) = 0.414253232

```
x2= 0.171006958
x3= 0.188985401
x1=  0.315911346
T by hand=703.965527
V by hand=-160.814156
m2=35.4
R=  0.0291099304
Energy without mass = 979.824899
Energy=  21099.8249
R=  0.0291099304
normx2 (computationally)=  -0.664963408
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305569
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.4346784
0 to x2 (h4 by hand)=  11.4346775
x2 to x3 (h5 computationally)=  2.29520218
x2 to x3 (h5 by hand)=  2.29520447
x2 to x3 (h5 by handdddd)=  2.29520324
x3 to x1 (h6 computationally)=  2.93061968
x3 to x1 (h6 by hand)=  2.93062042
K22  (computationally) = 0.784753727
K22  (by hand) = 0.784753308
K32  (computationally) = 0.742894316
K32  (by hand) = 0.742894001
K21  (computationally) = 0.479666506
K21  (by hand) =0.479666465
K31  (computationally) = 0.475362851
K31  (by hand) = 0.475363419
K33  (computationally) = 0.710886352
K33  (by hand) = 0.710888499
K11  (computationally) = 0.414860344
K11  (by hand) = 0.414861898
x2= 0.170761626
```

```
x3= 0.188713212
x1=  0.315443824
T by hand=706.040369
V by hand=-161.0483
m2=35.5
R=  0.0291099304
Energy without mass = 979.729745
Energy=  21099.7297
R=  0.0291099304
normx2 (computationally)=  -0.664963489
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305565
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.4674556
0 to x2 (h4 by hand)=  11.4674556
x2 to x3 (h5 computationally)=  2.30188195
x2 to x3 (h5 by hand)=  2.30188302
x2 to x3 (h5 by handdddd)=  2.30188301
x3 to x1 (h6 computationally)=  2.93934812
x3 to x1 (h6 by hand)=  2.93934881
K22  (computationally) = 0.785872863
K22  (by hand) = 0.785872877
K32  (computationally) = 0.74395507
K32  (by hand) = 0.74395508
K21  (computationally) = 0.480363239
K21  (by hand) =0.48036324
K31  (computationally) = 0.476053143
K31  (by hand) = 0.476053424
K33  (computationally) = 0.711900727
K33  (by hand) = 0.711901847
K11  (computationally) = 0.415467313
K11  (by hand) = 0.415468704
x2= 0.170517355
x3= 0.188442174
```

```
x1=  0.31497833
T by hand=708.115213
V by hand=-161.281873
m2=35.6
R=  0.0291099304
Energy without mass = 979.649257
Energy=  21099.6493
R=  0.0291099304
normx2 (computationally)=  -0.664963489
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305563
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.5002322
0 to x2 (h4 by hand)=  11.5002322
x2 to x3 (h5 computationally)=  2.30856395
x2 to x3 (h5 by hand)=  2.30856503
x2 to x3 (h5 by handdddd)=  2.30856502
x3 to x1 (h6 computationally)=  2.94807724
x3 to x1 (h6 by hand)=  2.94807791
K22  (computationally) = 0.786989658
K22  (by hand) = 0.786989668
K32  (computationally) = 0.745013889
K32  (by hand) = 0.745013896
K21  (computationally) = 0.481058944
K21  (by hand) =0.481058945
K31  (computationally) = 0.476742893
K31  (by hand) = 0.476743178
K33  (computationally) = 0.712915237
K33  (by hand) = 0.71291637
K11  (computationally) = 0.416073603
K11  (by hand) = 0.416074954
x2= 0.170274115
x3= 0.188172301
x1=  0.314514866
```

```
T by hand=710.190219
V by hand=-161.515219
m2=35.7
R=  0.0291099304
Energy without mass = 979.583171
Energy=  21099.5832
R=  0.0291099304
normx2 (computationally)=  -0.664963492
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305574
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.5330096
0 to x2 (h4 by hand)=  11.5330097
x2 to x3 (h5 computationally)=  2.31524616
x2 to x3 (h5 by hand)=  2.3152472
x2 to x3 (h5 by handdddd)=  2.31524723
x3 to x1 (h6 computationally)=  2.95680725
x3 to x1 (h6 by hand)=  2.95680806
K22  (computationally) = 0.788104923
K22  (by hand) = 0.788104949
K32  (computationally) = 0.746071244
K32  (by hand) = 0.746071264
K21  (computationally) = 0.481753702
K21  (by hand) =0.481753704
K31  (computationally) = 0.477431683
K31  (by hand) = 0.477431958
K33  (computationally) = 0.713928256
K33  (by hand) = 0.713929356
K11  (computationally) = 0.416679519
K11  (by hand) = 0.416681133
x2= 0.170031908
x3= 0.187903575
x1=  0.314053421
T by hand=712.265402
```

```
V by hand=-161.74834
m2=35.8
R=  0.0291099304
Energy without mass = 979.531375
Energy=  21099.5314
R=  0.0291099304
normx2 (computationally)=  -0.664963486
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305564
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.5657877
0 to x2 (h4 by hand)=  11.5657877
x2 to x3 (h5 computationally)=  2.32192888
x2 to x3 (h5 by hand)=  2.32193001
x2 to x3 (h5 by handdddd)=  2.32192996
x3 to x1 (h6 computationally)=  2.96553749
x3 to x1 (h6 by hand)=  2.96553819
K22  (computationally) = 0.789218547
K22  (by hand) = 0.789218543
K32  (computationally) = 0.747127075
K32  (by hand) = 0.747127071
K21  (computationally) = 0.482447439
K21  (by hand) =0.482447439
K31  (computationally) = 0.478119513
K31  (by hand) = 0.47811981
K33  (computationally) = 0.71494003
K33  (by hand) = 0.714941209
K11  (computationally) = 0.417283825
K11  (by hand) = 0.417285219
x2= 0.169790725
x3= 0.187635991
x1=  0.313593961
T by hand=714.340606
V by hand=-161.980973
```

```
m2=35.9
R=  0.0291099304
Energy without mass = 979.493753
Energy=  21099.4938
R=  0.0291099304
normx2 (computationally)=  -0.664963491
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305565
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.5985667
0 to x2 (h4 by hand)=  11.5985667
x2 to x3 (h5 computationally)=  2.32861163
x2 to x3 (h5 by hand)=  2.32861269
x2 to x3 (h5 by handdddd)=  2.32861271
x3 to x1 (h6 computationally)=  2.97426858
x3 to x1 (h6 by hand)=  2.97426929
K22  (computationally) = 0.790330724
K22  (by hand) = 0.790330747
K32  (computationally) = 0.748181495
K32  (by hand) = 0.748181512
K21  (computationally) = 0.483140237
K21  (by hand) =0.48314024
K31  (computationally) = 0.478806348
K31  (by hand) = 0.478806628
K33  (computationally) = 0.715950185
K33  (by hand) = 0.715951305
K11  (computationally) = 0.417887677
K11  (by hand) = 0.417889098
x2= 0.169550561
x3= 0.187369538
x1=  0.313136489
T by hand=716.415973
V by hand=-162.21336
m2=36.
```

```
R=  0.0291099304
Energy without mass = 979.470144
Energy=  21099.4701
R=  0.0291099304
normx2 (computationally)=  -0.664963491
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305574
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.6313464
0 to x2 (h4 by hand)=  11.6313465
x2 to x3 (h5 computationally)=  2.33529477
x2 to x3 (h5 by hand)=  2.33529583
x2 to x3 (h5 by handdddd)=  2.33529586
x3 to x1 (h6 computationally)=  2.98300045
x3 to x1 (h6 by hand)=  2.98300127
K22  (computationally) = 0.791441315
K22  (by hand) = 0.791441342
K32  (computationally) = 0.749234427
K32  (by hand) = 0.749234447
K21  (computationally) = 0.483832084
K21  (by hand) =0.483832087
K31  (computationally) = 0.479492265
K31  (by hand) = 0.479492544
K33  (computationally) = 0.716959021
K33  (by hand) = 0.716960137
K11  (computationally) = 0.41849097
K11  (by hand) = 0.418492597
x2= 0.169311408
x3= 0.18710421
x1=  0.312680989
T by hand=718.491492
V by hand=-162.445488
m2=36.1
R=  0.0291099304
```

```
Energy without mass = 979.460405
Energy=  21099.4604
R=  0.0291099304
normx2 (computationally)=  -0.664963491
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305569
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.6641269
0 to x2 (h4 by hand)=  11.6641269
x2 to x3 (h5 computationally)=  2.34197823
x2 to x3 (h5 by hand)=  2.34197929
x2 to x3 (h5 by handdddd)=  2.34197932
x3 to x1 (h6 computationally)=  2.9917327
x3 to x1 (h6 by hand)=  2.99173347
K22  (computationally) = 0.792550361
K22  (by hand) = 0.792550389
K32  (computationally) = 0.750285896
K32  (by hand) = 0.750285917
K21  (computationally) = 0.484522944
K21  (by hand) =0.484522947
K31  (computationally) = 0.480177208
K31  (by hand) = 0.480177488
K33  (computationally) = 0.717966474
K33  (by hand) = 0.717967596
K11  (computationally) = 0.419092952
K11  (by hand) = 0.419094485
x2= 0.169073257
x3= 0.186839999
x1=  0.312227436
T by hand=720.567066
V by hand=-162.677191
m2=36.2
R=  0.0291099304
Energy without mass = 979.464423
```

```
Energy=  21099.4644
R=  0.0291099304
normx2 (computationally)=  -0.664963487
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305563
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.6346243
0 to x2 (h4 by hand)=  11.6346243
x2 to x3 (h5 computationally)=  2.33596321
x2 to x3 (h5 by hand)=  2.33596433
x2 to x3 (h5 by handdddd)=  2.33596429
x3 to x1 (h6 computationally)=  2.98387345
x3 to x1 (h6 by hand)=  2.98387414
K22  (computationally) = 0.791552249
K22  (by hand) = 0.791552254
K32  (computationally) = 0.749339616
K32  (by hand) = 0.749339619
K21  (computationally) = 0.48390119
K21  (by hand) =0.483901191
K31  (computationally) = 0.479560804
K31  (by hand) = 0.479561098
K33  (computationally) = 0.71705991
K33  (by hand) = 0.71706108
K11  (computationally) = 0.418550843
K11  (by hand) = 0.418552231
x2= 0.169287547
x3= 0.187077739
x1=  0.312635541
T by hand=718.698999
V by hand=-162.468596
m2=36.11
R=  0.0291099304
Energy without mass = 979.460199
Energy=  21099.4602
```

```
R=  0.0291099304
normx2 (computationally)=  -0.664963491
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305574
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.6379024
0 to x2 (h4 by hand)=  11.6379025
x2 to x3 (h5 computationally)=  2.33663144
x2 to x3 (h5 by hand)=  2.3366325
x2 to x3 (h5 by handdddd)=  2.33663253
x3 to x1 (h6 computationally)=  2.98474688
x3 to x1 (h6 by hand)=  2.9847477
K22  (computationally) = 0.791663245
K22  (by hand) = 0.791663272
K32  (computationally) = 0.749444836
K32  (by hand) = 0.749444856
K21  (computationally) = 0.483970335
K21  (by hand) =0.483970338
K31  (computationally) = 0.479629333
K31  (by hand) = 0.479629613
K33  (computationally) = 0.717160627
K33  (by hand) = 0.717161746
K11  (computationally) = 0.41861148
K11  (by hand) = 0.418613117
x2= 0.169263697
x3= 0.187051279
x1=  0.312590123
T by hand=718.906604
V by hand=-162.491865
m2=36.12
R=  0.0291099304
Energy without mass = 979.460112
Energy=  21099.4601
R=  0.0291099304
```

```
normx2 (computationally)=  -0.664963439
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305569
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.6411796
0 to x2 (h4 by hand)=  11.6411791
x2 to x3 (h5 computationally)=  2.33730104
x2 to x3 (h5 by hand)=  2.3373029
x2 to x3 (h5 by handdddd)=  2.33730213
x3 to x1 (h6 computationally)=  2.98562
x3 to x1 (h6 by hand)=  2.98562076
K22  (computationally) = 0.79177369
K22  (by hand) = 0.791773434
K32  (computationally) = 0.749549729
K32  (by hand) = 0.749549537
K21  (computationally) = 0.484039403
K21  (by hand) =0.484039378
K31  (computationally) = 0.479698118
K31  (by hand) = 0.479698587
K33  (computationally) = 0.717262417
K33  (by hand) = 0.717264212
K11  (computationally) = 0.418671603
K11  (by hand) = 0.418673177
x2= 0.169239853
x3= 0.187024835
x1=  0.312544719
T by hand=719.11416
V by hand=-162.515063
m2=36.13
R=  0.0291099304
Energy without mass = 979.460152
Energy=  21099.4602
R=  0.0291099304
normx2 (computationally)=  -0.664963491
```

```
normx2 (by hand)=  -0.664963533
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305565
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.6382302
0 to x2 (h4 by hand)=  11.6382303
x2 to x3 (h5 computationally)=  2.33669828
x2 to x3 (h5 by hand)=  2.33669934
x2 to x3 (h5 by handdddd)=  2.33669937
x3 to x1 (h6 computationally)=  2.98483402
x3 to x1 (h6 by hand)=  2.98483474
K22  (computationally) = 0.791674341
K22  (by hand) = 0.791674367
K32  (computationally) = 0.749455355
K32  (by hand) = 0.749455375
K21  (computationally) = 0.483977229
K21  (by hand) =0.483977231
K31  (computationally) = 0.479636167
K31  (by hand) = 0.479636447
K33  (computationally) = 0.717170705
K33  (by hand) = 0.717171824
K11  (computationally) = 0.418617174
K11  (by hand) = 0.418618602
x2= 0.169261313
x3= 0.187048634
x1=  0.312585577
T by hand=718.927317
V by hand=-162.494111
m2=36.121
R=  0.0291099304
Energy without mass = 979.460122
Energy=  21099.4601
R=  0.0291099304
normx2 (computationally)=  -0.664963486
normx2 (by hand)=  -0.664963533
```

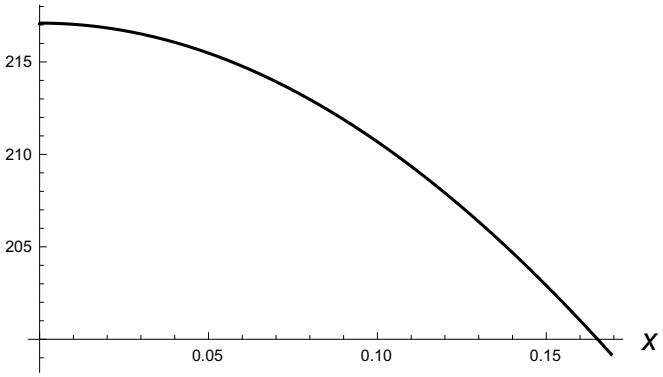
```
normx3 (computationally)=  -0.628678171
normx3 (by hand)=  -0.628678207
normx1 (computationally)=  -0.397305562
normx1 (by hand)=  -0.397305574
0 to x2 (h4 computationally)=  11.6385579
0 to x2 (h4 by hand)=  11.6385579
x2 to x3 (h5 computationally)=  2.33676523
x2 to x3 (h5 by hand)=  2.33676636
x2 to x3 (h5 by handdddd)=  2.33676632
x3 to x1 (h6 computationally)=  2.98492128
x3 to x1 (h6 by hand)=  2.98492195
K22  (computationally) = 0.791685389
K22  (by hand) = 0.79168539
K32  (computationally) = 0.749465847
K32  (by hand) = 0.749465847
K21  (computationally) = 0.48398413
K21  (by hand) =0.48398413
K31  (computationally) = 0.479643038
K31  (by hand) = 0.479643335
K33  (computationally) = 0.717180877
K33  (by hand) = 0.717182058
K11  (computationally) = 0.418623076
K11  (by hand) = 0.418624427
x2= 0.169258928
x3= 0.187045989
x1=  0.312581034
T by hand=718.948059
V by hand=-162.496406
m2=36.122
R=  0.0291099304
Energy without mass = 979.460124
Energy=  21099.4601
x1=0.312581034
x3=0.187045989
x2=0.169258928
Kinetic Energy=718.948059
```

Potential Energy=-162.496406

Volume Energy=423.008472

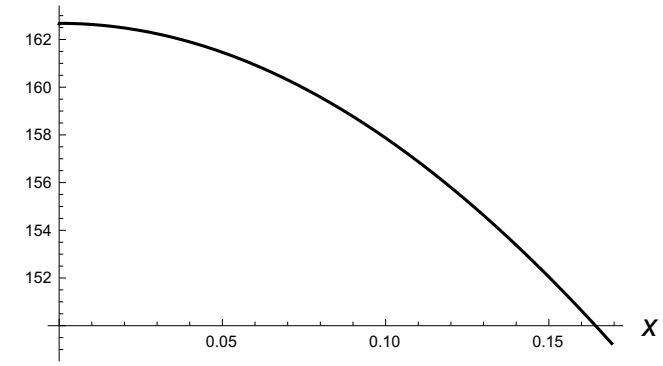
$(f_2(x)/x)^{3/2}$

Out[ ]=



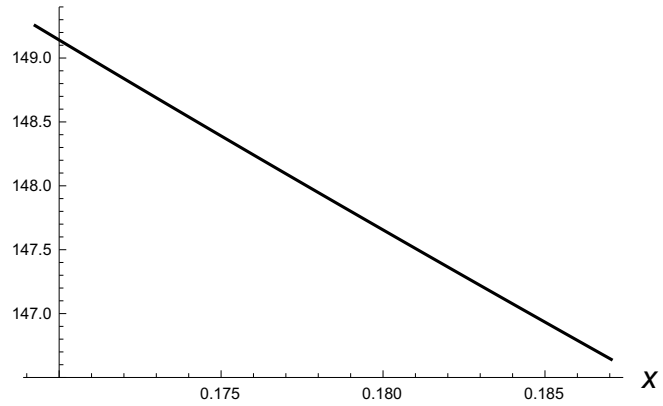
$(f_{2\,bar}(x)/x)^{3/2}$

Out[ ]=

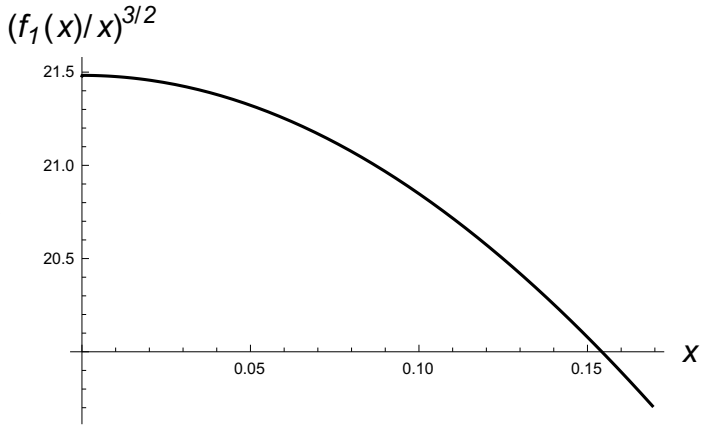


$(f_{2\,bar}(x)/x)^{3/2}$

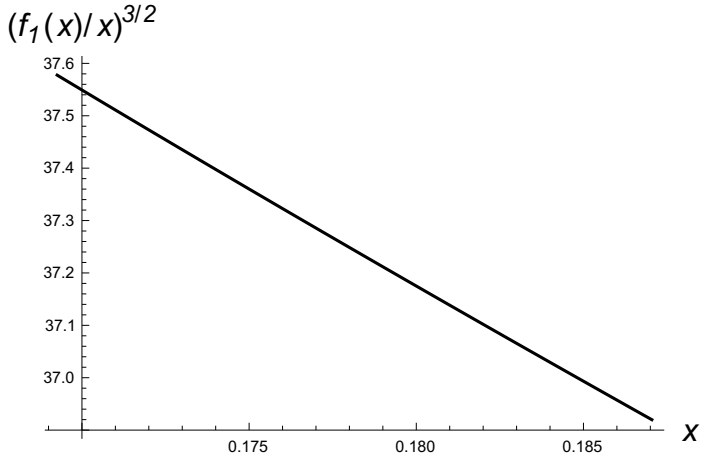
Out[ ]=



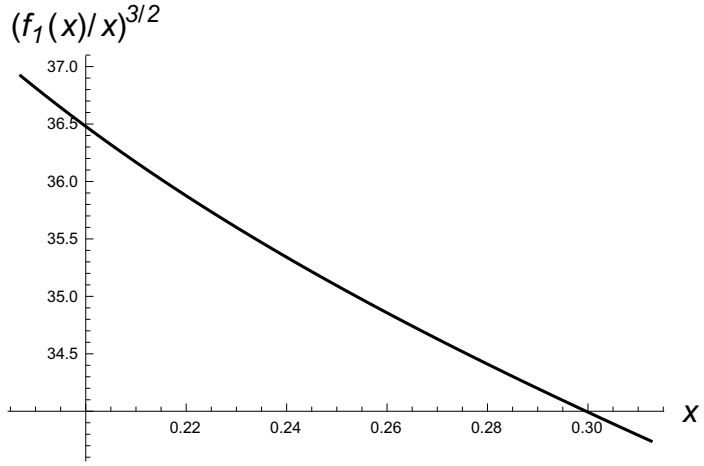
Out[ ]=


$(f_1(x)/x)^{3/2}$
21.5
21.0
20.5
0.05
0.10
0.15
x

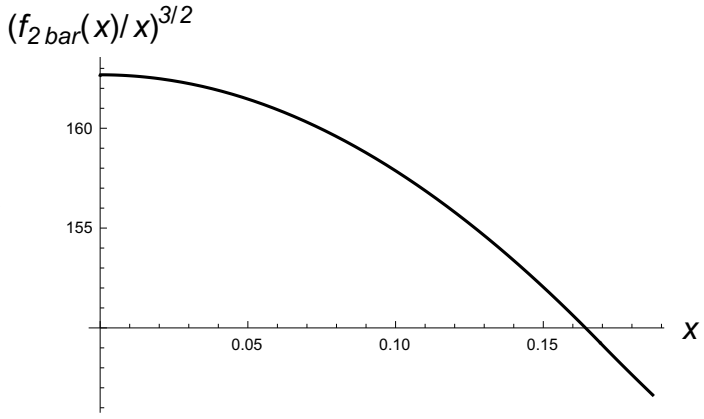
Out[ ]=


$(f_1(x)/x)^{3/2}$
37.6
37.5
37.4
37.3
37.2
37.1
37.0
0.175
0.180
0.185
x

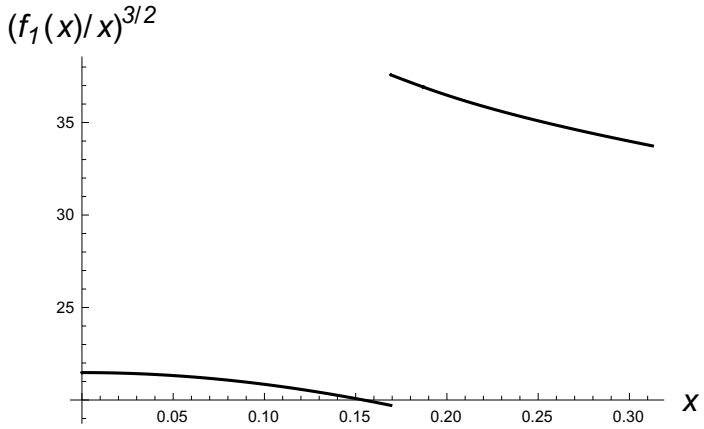
Out[ ]=


$(f_1(x)/x)^{3/2}$
37.0
36.5
36.0
35.5
35.0
34.5
0.22
0.24
0.26
0.28
0.30
x

Out[ ]=


$(f_{2\,bar}(x)/x)^{3/2}$
160
155
0.05
0.10
0.15
x

Out[ ]=


$(f_1(x)/x)^{3/2}$
35
30
25
0.05
0.10
0.15
0.20
0.25
0.30
x